

$$E = 200 \text{ GPa}$$

$$I = 108 \times 10^6 \text{ mm}^4$$

$$[K] \begin{Bmatrix} 0 \\ 0 \\ 0 \\ u_4 \\ 0 \\ u_6 \\ 0 \\ 0 \end{Bmatrix} = \begin{Bmatrix} R_1 \\ R_2 \\ R_3 \\ F_4 \\ R_5 \\ F_6 \\ R_7 \\ R_8 \end{Bmatrix}$$

$$\text{DOF}(j, u) = 2 \times 4 - 1 = 7$$

$$\text{DOF}(j, \theta) = 2j = 8$$

Since element 1 and 3 have same $L = 4 \text{ m}$,

$$\frac{EI}{L^3} = \frac{200 \times 108 \times 10^6 \times 10^9}{4000^3} = 337.5 \text{ kN/m}$$

$$\frac{\text{kN}}{\text{mm}^2 \text{ mm}^3} = \frac{\text{m}^2}{\text{m}^4} = \frac{1}{\text{m}^2}$$

$$K_3 = K_1 = \frac{EI}{L^3} \begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}$$

$$= 337.5 \begin{bmatrix} 12 & 6 \times 4 & -12 & 6 \times 4 \\ 6 \times 4 & 4 \times 4^2 & -6 \times 4 & 2 \times 4^2 \\ -12 & -6 \times 4 & 12 & -6 \times 4 \\ -6 \times 4 & -4 \times 4^2 & -6 \times 4 & -6 \times 4 \end{bmatrix}$$

$$= \begin{bmatrix} 4050 & 8100 & -4050 & 8100 \\ 8100 & 21600 & -8100 & 10800 \\ -4050 & -8100 & 4050 & -8100 \\ 8100 & -10800 & -8100 & 21600 \end{bmatrix}$$

Similarly,
for element 2,

$$L = 3m$$

$$\therefore \frac{EI}{L^3} = \frac{200 \times 10^8 \times 10^6 \times 10^3}{33 \times 1000^3} = 800 \text{ kN/m}$$

$$\therefore k_2 = \frac{EI}{L^3} \begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}$$

$$= \begin{bmatrix} 9600 & 14400 & -9600 & 14400 \\ 14400 & 28800 & -14400 & 14400 \\ -9600 & -14400 & 9600 & -14400 \\ 14400 & 14400 & 14400 & 28800 \end{bmatrix}$$

element 1 $\rightarrow$ local (1, 2, 3, 4) $\rightarrow$ global (1, 2, 3, 4)

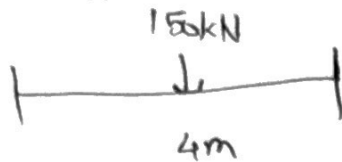
" 2 $\rightarrow$ " " $\rightarrow$ " (3, 4, 5, 6)

" 3 $\rightarrow$ " " $\rightarrow$ " (5, 6, 7, 8)

$\therefore$ Assemble global $k =$

	1	2	3	4	5	6	7	8
1	9600	8100	-4050	8100	0	0	0	0
2	8100	21600	-8100	10800	0	0	0	0
3	-4050	-8100	13650	6300	-9600	14400	0	0
4	8100	10800	6300	50400	-14400	14400	0	0
5	0	0	-9600	-14400	13650	-6300	-4500	8100
6	0	0	14400	14400	-6300	50400	-8100	10800
7	0	0	0	0	-4500	-8100	4050	-8100
8	0	0	0	0	8100	10800	-8100	21600

For element 1,



For element 2

$$Q_{f1} = \begin{bmatrix} P/2 \\ PL/8 \\ P/2 \\ -PL/8 \end{bmatrix} = \begin{bmatrix} 75 \\ 75 \\ 75 \\ -75 \end{bmatrix}$$

$$Q_{f2} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

For element 3,

$$Q_{f3} = \begin{bmatrix} qL/2 \\ qL^2/12 \\ qL/2 \\ -qL^2/12 \end{bmatrix} = \begin{bmatrix} 37.5 \times 4/2 \\ 37.5 \times 4^2/12 \\ 37.5 \times 4/2 \\ -37.5 \times 4^2/12 \end{bmatrix} = \begin{bmatrix} 75 \\ 50 \\ 75 \\ -50 \end{bmatrix}$$

$$\therefore F_F = \begin{bmatrix} 75 \\ 75 \\ 75 \\ -75 \\ 75 \\ 50 \\ 75 \\ -50 \end{bmatrix}$$

Free DOF = 4, 6

Restrained = 1, 2, 3, 5, 7, 8

$$U_R = 0 \quad F_F = 0$$

$$\therefore U_f = K_{ff}^{-1} (F_f - F_f^F - K_{fr} U_R) = K_{ff}^{-1} F_f^F$$

$$K_{ff} = \begin{bmatrix} 50400 & 14400 \\ 4400 & 50400 \end{bmatrix}$$

$$F_f^F = \begin{bmatrix} -75 \\ 50 \end{bmatrix}$$

$$K_{ff}^{-1} = \begin{bmatrix} .000021 & -.000006 \\ -.000006 & .000021 \end{bmatrix}$$

$$-K_{ff}^{-1} F_f^F = U_f$$

$$U_f = \begin{bmatrix} .0019 \\ -.00154 \end{bmatrix}$$

$$\therefore U = \begin{bmatrix} 0 \\ 0 \\ 0 \\ .001029 \\ 0 \\ -.00154 \\ 0 \\ 0 \end{bmatrix}$$

(3)

Recover support reactions:

$$F_R = k_{FF} u_F + k_{FR} u_R + F_R$$

$$= \begin{bmatrix} 8100 & 0 \\ 10800 & 0 \\ 6800 & 14400 \\ -14400 & -6800 \\ 0 & -8100 \\ 0 & 10800 \end{bmatrix} \times \begin{bmatrix} 0.0019 \\ -0.00154 \end{bmatrix} + \begin{bmatrix} 75 \\ 75 \\ 75 \\ 75 \\ 75 \\ -50 \end{bmatrix}$$

$$= \begin{bmatrix} 90.625 \\ 95.833 \\ 64.931 \\ 56.944 \\ 87.5 \\ -66.67 \end{bmatrix}$$

$$\therefore V_1 = 90.625 \text{ kN}$$

$$M_1 = 95.833 \text{ kNm}$$

$$V_2 = 64.931 \text{ kN}$$

~~M2 =~~

$$V_3 = 56.944 \text{ kN}$$

$$V_4 = 87.5 \text{ kN}$$

$$M_4 = -66.67 \text{ kN-m}$$

Global equilibrium check:

$$V_1 + V_2 + V_3 + V_4 = 90.625 + 64.931 + 56.944 + 87.5 = 300$$

$$\text{Total external force} = -150 - 37.5 \times 4 = -300$$

$\therefore$ OK.

Local element end displacement: Local end force.

② Element 1:

$$\begin{bmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0.0019 \end{bmatrix}$$

~~Element 2~~

$$Q(1) = k^1 u^1 + Q_f^1 = \begin{bmatrix} 4050 & 8100 & -4050 & 8100 \\ 8100 & 21600 & -8100 & 10800 \\ -4050 & -8100 & 4050 & -8100 \\ 8100 & 10800 & -8100 & 21600 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0.0019 \end{bmatrix} + \begin{bmatrix} 75 \\ 75 \\ 75 \\ -75 \end{bmatrix} = \begin{bmatrix} 90.625 \\ 95.83 \\ 50.37 \\ -33.33 \end{bmatrix}$$

∴ For element 1,

$$V_b = 90.625 \text{ kN}$$

$$M_b = 95.833 \text{ kNm}$$

$$V_e = 50.37 \text{ kN}$$

$$M_e = -33.33 \text{ kNm}$$

Similarly,

$$Q(2) = k^2 u^2 + Q_f^2 = \begin{bmatrix} 0600 & 14400 & -0600 & 14400 \\ 14400 & 28800 & -14400 & 14400 \\ -0600 & -14400 & 0600 & -14400 \\ 14400 & 14400 & 14400 & 28800 \end{bmatrix} \begin{bmatrix} 0 \\ 0.0019 \\ 0 \\ -0.00154 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\therefore Q(2) = \begin{bmatrix} 5.56 \\ 33.33 \\ -5.56 \\ -16.67 \end{bmatrix}$$

For element 2,

$$V_b = 5.56 \text{ kN}$$

$$M_b = 3.33 \text{ kNm}$$

$$V_e = -5.56 \text{ kN}$$

$$M_e = -16.67 \text{ kNm}$$

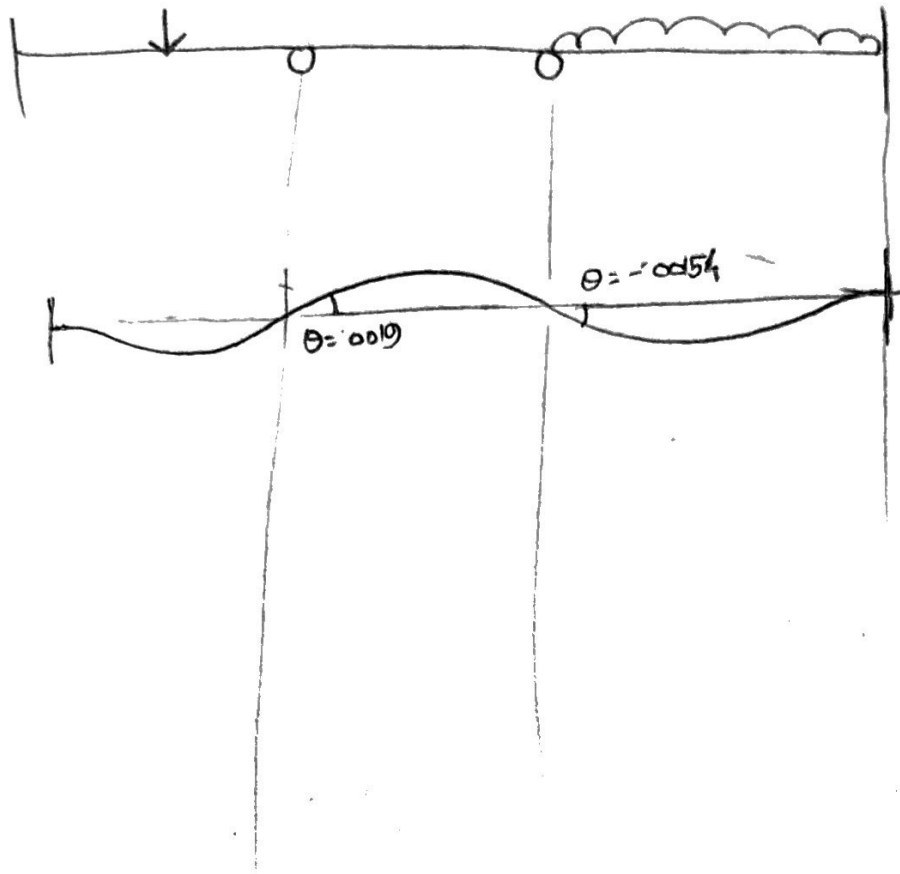
$$Q(3) = \begin{bmatrix} 4050 & 8100 & -4050 & 8100 \\ 8100 & 21600 & -8100 & 10800 \\ -4050 & -8100 & 4050 & -8100 \\ 8100 & 10800 & -8100 & 21600 \end{bmatrix} \begin{bmatrix} 0 \\ -0.00154 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 75 \\ 50 \\ 75 \\ -50 \end{bmatrix} = \begin{bmatrix} 62.5 \\ 16.67 \\ 87.5 \\ -66.67 \end{bmatrix}$$

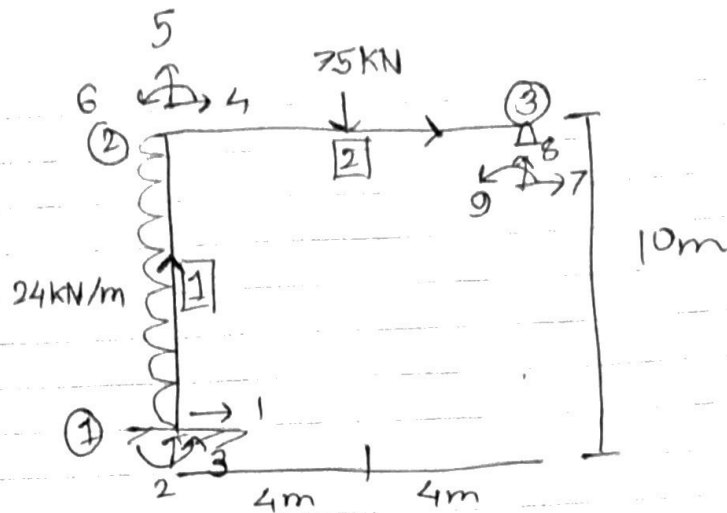
$$V_b = 62.5 \text{ kN}$$

$$M_b = 66.67 \text{ kN-m}$$

$$V_e = 87.5 \text{ kN}$$

$$M_e = -66.67 \text{ kN-m}$$





$$E = 200 \text{ GPa} = 2 \times 10^8 \text{ kN/m}^2$$

$$A = 4740 \text{ mm}^2 = 4740 \times 10^{-6} \text{ m}^2$$

$$I = 22.2 \times 10^6 \text{ mm}^4 = 22.2 \times 10^{-6} \text{ m}^4$$

$$1 = u_{1x}$$

$$2 = u_{1y}$$

$$3 = \theta_1$$

$$4 = u_{2x}$$

$$5 = u_{2y}$$

$$6 = \theta_2$$

$$7 = u_{3x}$$

$$8 = u_{3y}$$

$$9 = \theta_3$$

Global applied load

$$F = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{Bmatrix}$$

Restraints

Node 1 fixed

$$\therefore u_{1x}, u_{1y}, \theta_1 = 0$$

Node 3 is pin

$$\therefore u_{3x}, u_{3y} = 0$$

$$\therefore \text{restrained DOF} = 1, 2, 3, 7, 8$$

$$\text{free DOF} = 4, 5, 6, 9$$

For element 1,

$$L = 10m$$

$$\frac{EI}{L^3} = \frac{200 \times 10^6 \times 22.2 \times 10^6}{10^{12} \times 10^3} = 4.44$$

$$\frac{AL^2}{I} = \frac{4740 \times 10^{-6} \times 10^2 \times 10^{12}}{22.2 \times 10^6} = 21351.35$$

$$6L = 6 \times 10 = 60$$

$$4L^2 = 4 \times 10^2 = 400$$

$$2L^2 = 2 \times 10^2 = 200$$

$$\therefore k^4 = \frac{EI}{L^3} \begin{bmatrix} \frac{AL^2}{I} & 0 & 0 & -\frac{AL^2}{I} & 0 & 0 \\ 0 & 12 & 6L & 0 & -12 & 6L \\ 0 & 6L & 4L^2 & 0 & -6L & 2L^2 \\ -\frac{AL^2}{I} & 0 & 0 & \frac{AL^2}{I} & 0 & 0 \\ 0 & -12 & -6L & 0 & 12 & -6L \\ 0 & 6L & 2L^2 & 0 & -6L & 4L^2 \end{bmatrix}$$

$$= 4.44 \begin{bmatrix} 21351.35 & 0 & 0 & -21351 & 0 & 0 \\ 0 & 12 & 60 & 0 & -12 & 60 \\ 0 & 60 & 400 & 0 & -60 & 200 \\ -21351.35 & 0 & 0 & 21351 & 0 & 0 \\ 0 & -12 & -60 & 0 & 12 & -60 \\ 0 & 60 & 200 & 0 & -60 & 400 \end{bmatrix}$$

$$= \begin{bmatrix} 94800 & 0 & 0 & -94800 & 0 & 0 \\ 0 & 53.28 & 266.4 & 0 & -53.28 & 266.4 \\ 0 & 266.4 & 1776 & 0 & -266.4 & 888 \\ -94800 & 0 & 0 & 94800 & 0 & 0 \\ 0 & -53.28 & -266.4 & 0 & 53.28 & -266.4 \\ 0 & 266.4 & 888 & 0 & -266.4 & 1776 \end{bmatrix}$$

similarly,

$$\frac{EI}{L^3} = \frac{2 \times 10^8 \times 22.2 \times 10^{-6}}{8^3} = 867$$

$$\frac{AL^2}{I} = \frac{4740 \times 10^{-6} \times 8^2}{22.2 \times 10^{-6}} = 1366.89$$

$$6L = 6 \times 8 = 48$$

$$4L^2 = 4 \times 8^2 = 256$$

$$2L^2 = 2 \times 8^2 = 128$$

$$\therefore k^2 \begin{bmatrix} 118500 & 0 & 0 & -118500 & 0 \\ 0 & 10406 & 416.25 & 0 & -10406 \\ 0 & 416.25 & 2220 & 0 & -416.25 \\ -118500 & 0 & 0 & 118500 & 0 \\ 0 & -10406 & -416.25 & 0 & 10406 \\ 0 & 416.25 & 1110 & 0 & -416.25 \end{bmatrix}$$

for element 1,

$$\theta = 90^\circ$$

$$c = \cos 90^\circ = 0$$

$$s = \sin 90^\circ = 1$$

$$\therefore T^{(1)} = \begin{bmatrix} c & s & 0 & 0 & 0 & 0 \\ -s & c & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$(I^1)^T = \begin{bmatrix} 0 & -1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$K^1 = T^T K T$$

$$= \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} K^1 T$$

$$= \begin{bmatrix} 5328 & 0 & -266.4 & 53.28 & 0 & -266.4 \\ 0 & 94800 & 0 & 0 & -94800 & 0 \\ -266.4 & 0 & 1776 & 266.4 & 0 & 888 \\ -53.28 & 0 & 266.4 & 53.28 & 0 & 266.4 \\ 0 & -94800 & 0 & 0 & 94800 & 0 \\ -266.4 & 0 & 888 & 266.4 & 0 & 1776 \end{bmatrix}$$

Since element 2 is horizontal.

$$\theta = 0 \\ c = 1 \\ s = 0$$

$$T = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \therefore T^T = T$$

$$\therefore K^2 = K^1$$

$$\therefore K = \begin{bmatrix} 53.28 & 0 & -266.4 & 53.28 & 0 & -266.4 & 0 & 0 & 0 \\ 0 & 94800 & 0 & 0 & -94800 & 0 & 0 & 0 & 0 \\ -266.4 & 0 & 1776 & 266.4 & 0 & 888 & 0 & 0 & 0 \\ -53.28 & 0 & 266.4 & 53.28 & 0 & 266.4 & -118500 & 0 & 0 \\ 0 & -94800 & 0 & 0 & 94800 & 416.3 & 0 & -104 & 416.3 \\ -266.4 & 0 & 888 & 266.4 & 416.3 & 3006 & 0 & -416.3 & 1110 \\ 0 & 0 & 0 & -118500 & 0 & 0 & -118500 & 0 & 0 \\ 0 & 0 & 0 & 0 & -104 & -416.3 & 0 & 104 & -416.3 \\ 0 & 0 & 0 & 0 & 416.3 & 1110 & 0 & -416.3 & 2220 \end{bmatrix}$$

For element 1, $udl = 24 \text{ kNm}^{-1}$

$$Q_f^{(1)} = \begin{bmatrix} 0 \\ -qL/2 \\ -qL^2/12 \\ 0 \\ -qL/2 \\ qL^2/12 \end{bmatrix} = \begin{bmatrix} 0 \\ -120 \\ -200 \\ 0 \\ -120 \\ 200 \end{bmatrix}$$

$$qL/2 = \frac{24 \times 10}{2} = 120$$

$$qL^2/12 = \frac{24 \times 10^2}{12} = 200$$

$$F_f = T^T Q_f = \begin{bmatrix} -120 \\ 0 \\ +200 \\ -120 \\ 0 \\ -200 \end{bmatrix}$$

For element 2, point load = -75 kN

$$Q_f^{(2)} = \begin{bmatrix} 0 \\ -P/2 \\ -PL/8 \\ 0 \\ -P/2 \\ +PL/8 \end{bmatrix} = \begin{bmatrix} 0 \\ +37.5 \\ +75 \\ 0 \\ +37.5 \\ -75 \end{bmatrix} = F_f^{(2)}$$

$$P/2 = \frac{75}{2} = 37.5$$

$$PL/8 = \frac{75 \times 8}{8} = 75$$

Since $T^T = I$

$$F_f^{(2)} = Q_f^{(2)}$$

For element 1, Local DOF (1, 2, 3, 4, 5, 6) $\rightarrow$ G (2, 1, 3, 5, 4, 6)
 " " 2, Local DOF (1, 2, 3, 4, 5, 6) $\rightarrow$ G (4, 5, 6, 7, 8, 9)

$$\therefore F^F = \begin{bmatrix} -120 \\ 0 \\ 200 \\ -120 \\ 37.5 \\ -12.5 \\ 0 \\ 37.5 \\ -7.5 \end{bmatrix}$$

Now,

$$U_{FE} = 0$$

$F^F = 0$ as no nodal loads are applied.

$$\therefore K U = F - F^F$$

$$\text{or, } K U = -F^F$$

$$\text{now, } U_{FE} = \begin{bmatrix} U_{1x} \\ U_{1y} \\ \theta_1 \\ U_{3x} \\ U_{3y} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\therefore U_f = \begin{bmatrix} U_{2x} \\ U_{2y} \\ \theta_2 \\ \theta_3 \end{bmatrix}$$

$$\therefore K_f U_f = F_f$$

$$K_f U_f = \begin{bmatrix} 11855.3 & 0 & 266.4 & 0 \\ 0 & 9490.4 & 416.25 & 416.25 \\ 266.4 & 416.25 & 3006 & 110 \\ 0 & 416.25 & 110 & 2220 \end{bmatrix} \cdot \begin{bmatrix} U_{2x} \\ U_{2y} \\ \theta_2 \\ \theta_3 \end{bmatrix} = \begin{bmatrix} 120 \\ -37.5 \\ 12.5 \\ 7.5 \end{bmatrix}$$

$$\therefore U_f = \begin{bmatrix} U_{2x} \\ U_{2y} \\ \theta_2 \\ \theta_3 \end{bmatrix} = \begin{bmatrix} 0.0006 \\ -0.0006 \\ 0.0254 \\ 0.0212 \end{bmatrix}$$

$$U = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0.0006 \\ -0.0006 \\ 0.0254 \\ 0 \\ 0 \\ 0.0212 \end{bmatrix}$$

$$F_R = K_{RF} U_f + F_R^F$$

$$U_f = \begin{bmatrix} U_{1x} \\ U_{1y} \\ \theta_1 \\ U_{3x} \\ U_{3y} \end{bmatrix}$$

$$\therefore F_R = \begin{bmatrix} -53.28 & 0 & -266.4 & 0 \\ 0 & -94800 & 0 & 0 \\ 266.4 & 0 & 888 & 0 \\ -118500 & 0 & 0 & 0 \\ 0 & -104.06 & -416.25 & -416.25 \end{bmatrix} \begin{bmatrix} 0.0006 \\ -0.0006 \\ 0.0254 \\ 0.0212 \end{bmatrix} +$$

$$\begin{bmatrix} -120 \\ 0 \\ 200 \\ 0 \\ 37.5 \end{bmatrix}$$

$$\therefore F_R = \begin{bmatrix} -68.2 \\ 56.83 \\ 22.8 \\ -113.19 \\ -10.33 \end{bmatrix} + \begin{bmatrix} -120 \\ 0 \\ 200 \\ 0 \\ 37.5 \end{bmatrix} = \begin{bmatrix} -126.8 \\ 56.83 \\ 222.8 \\ -113.2 \\ 18.2 \end{bmatrix}$$

$$\begin{aligned}\therefore R_{1x} &= -126.82 \text{ kN} \\ R_{1y} &= 56.83 \text{ kN} \\ M_1 &= 222.8 \text{ kN-m} \\ R_{3x} &= -113.18 \text{ kN} \\ R_{3y} &= 18.17 \text{ kN}\end{aligned}$$

Force equilb. check:

$$\sum F_x = -126.82 + 113.18 + 24 \times 10 = 0 \text{ (OK)}$$

$$\sum F_y = 56.83 + 18.17 - 75 = 0 \text{ (OK)}$$

$$Q' = k' u' + Q_f'$$

$$u' = T' u = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0.00006 \\ -0.0006 \\ 0.254 \end{bmatrix}$$

$$u' = \begin{bmatrix} 0 \\ 0 \\ 0 \\ -0.0006 \\ -0.0006 \\ 0.254 \end{bmatrix}$$

$$\therefore Q' = \begin{bmatrix} 04800 & 0 & 0 & -04800 \\ 0 & 53.28 & 266.4 & 0 \\ 0 & 266.4 & 1776 & 0 \\ -04800 & 0 & 0 & 04800 \\ 0 & -53.28 & -266.4 & 0 \\ 0 & 266.4 & 888 & 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 120 \\ 280 \\ 120 \\ -200 \end{bmatrix} = \begin{bmatrix} 56.83 \\ 126.82 \\ 222.8 \\ -56.83 \\ 113.18 \\ -154.65 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 0 \\ 0 \\ -0.0006 \\ -0.0006 \\ 0.254 \end{bmatrix}$$

Similarly:
for element 2.

$$U_2 = V_2$$

$$\therefore Q_2^2 = k_2 \begin{bmatrix} 0.00096 & -0.0006 & 0.0254 & 0 & -0.0212 \\ -0.0006 & 0.0006 & 0 & 0 & 0 \\ 0.0254 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 37.5 & -7.5 \\ -0.0212 & 0 & 0 & -7.5 & 18.75 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 37.5 & 0 & 0 & 0 & 0 \\ 0 & 7.5 & 0 & 0 & 0 \\ 0 & 0 & 0 & 37.5 & -7.5 \\ 0 & 0 & 0 & -7.5 & 18.75 \end{bmatrix} = \begin{bmatrix} 113.19 & 0 & 0 & 0 & 0 \\ 56.83 & 0 & 0 & 0 & 0 \\ 154.65 & 0 & 0 & 0 & 0 \\ -113.19 & 0 & 0 & 0 & 0 \\ 18.17 & 0 & 0 & 0 & 0 \end{bmatrix}$$

